DSC 210, Fall 2026 — Homework 1 Nicholas Wendt (200478549)

Question 1: Property of triangular matrices (20 points)

Given U_1 and U_2 are two upper triangular matrices of size $n \times n$, prove that U_1U_2 is also an upper triangular matrix. Further, prove by induction that multiplication of $m\,(m > 2)$ upper triangular matrices $(U_1, U_2, \ldots, U_m)$ is also an upper triangular matrix.

Solution:

Since both U_1 and U_2 are upper triangular matrices that means entries $a_{ij} = 0$ for $i > j$ in U_1 and entries $b_{ij} = 0$ for $i > j$ in U_2.
Let matrix $C = U_1U_2$. We want to show that entries $c_{ij} = 0$ when $i > j$.
The element

$$c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

To show that every term in this sum is 0, we look at 2 cases.
Case 1: Let $k < i$, then element $a_{ik} = 0$
Case 2: Let $k \geq i$, then $k \geq i > j$, so $k > j$. Then element $b_{kj} = 0$
So $\forall k$, $a_{ik}b_{kj} = 0$

Therefore $c_{ij} = 0$ for $i > j$ making C an upper triangular matrix.

Inductive step: Assume $P_m = U_1U_2 \cdots U_m$ is an upper triangular matrix.
Let U_{m+1} be an upper triangular $n \times n$ matrix. Since matrix multiplication is associative,

$$P_{m+1} = U_1U_2 \cdots U_mU_{m+1} = (U_1U_2 \cdots U_m)\,U_{m+1} = P_m\,U_{m+1}.$$

By the inductive hypothesis, P_m is upper triangular, and U_{m+1} is upper triangular by definition. By the base case, the product of two upper triangular matrices is upper triangular.

So P_mU_{m+1} is upper triangular. Therefore P_{m+1} is upper triangular.

Conclusion: By induction, the product $U_1U_2 \cdots U_m$ of upper triangular matrices is an upper triangular matrix for all $m \geq 2$. ■

Question 2: Matrix operations (20 points)

Let $\mathbf{B}$ be a 4×4 matrix to which we apply the following 7 operations sequentially and get a final matrix $\mathbf{D}$:

(i) triple column 2,

(ii) halve row 4,

(iii) add row 2 to row 1,

(iv) interchange columns 1 and 3,

(v) subtract row 3 from each of the other rows,

(vi) replace column 4 by column 2,

(vii) delete column 1 (so that the column dimension is reduced by 1).

(a) Express each operation (i) to (vii) as a matrix and the final matrix $\mathbf{D}$ as a product of 8 matrices. (10 points)

(b) Write the final result again as a product of $\mathbf{ABC}$, i.e. write matrix $\mathbf{D} = \mathbf{ABC}$ and find $\mathbf{A}, \mathbf{C}$. (5 points)

(c) Write Python code to verify your answers in parts a and b. Show the answers and code. (5 points) Let

$$\mathbf{B} = \begin{bmatrix} 2 & 1 & 0 & 3 \\ 0 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

Hint: You can use NumPy for matrix operations.

Solution:
(a)

First let's write out the individual operations in matrix form.
Note: row operations must be multiplied on the left of B while column operations are multiplied on the right.

$$(i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, (ii) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}, (iii) = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, (iv) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(v) = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, (vi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, (vii) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now define matrix D:

$$D = (v)(iii)(ii)B(i)(iv)(vi)(vii)$$

(b) Let A be the product of row operations and let C be the product of column operations.

$$A = (v)(iii)(ii) = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

Multiplying right to left we get

$$A = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$$

Calculating C

$$C = (i)(iv)(vi)(vii) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Now we can write matrix D as the product of 3 matrices.

$$D = ABC = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 & 3 \\ 0 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(c)

```
import numpy as np
B = np.array([[2,1,0,3],[0,1,2,-1],[1,0,2,1],[0,1,1,2]])

i = np.array([[1,0,0,0],[0,3,0,0],[0,0,1,0],[0,0,0,1]])
ii = np.array([[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,0.5]])
iii = np.array([[1,1,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1]])
iv = np.array([[0,0,1,0],[0,1,0,0],[1,0,0,0],[0,0,0,1]])
```

```
v = np.array([[1,0,-1,0],[0,1,-1,0],[0,0,1,0],[0,0,-1,1]])
vi = np.array([[1,0,0,0],[0,1,0,1],[0,0,1,0],[0,0,0,0]])
vii = np.array([[0,0,0],[1,0,0],[0,1,0],[0,0,1]])

# part a
D = v @ iii @ ii @ B @ i @ iv @ vi @ vii
print("D =\n", D)

# part b
A = v @ iii @ ii
C = i @ iv @ vi @ vii
print("A =\n", A)
print("C =\n", C)
print("ABC =\n", A @ B @ C)
```

We see that

$$D = \begin{bmatrix} 6 & 1 & 6 \\ 3 & -1 & 3 \\ 0 & 1 & 0 \\ \frac{3}{2} & -1 & \frac{3}{2} \end{bmatrix}$$

Question 3: Matrix properties (20 points)

Prove that if a matrix $\mathbf{A}$ is triangular (upper or lower) then $\mathbf{A}^{-1}$ is also triangular. Further, use the result to show that if $\mathbf{A}$ is both triangular and orthogonal, then it is diagonal.

Solution:

Let A be an invertible upper triangular matrix.

Since $\det A$ is the product of the diagonal entries and $\det A \neq 0$, every diagonal entry a_{ii} is nonzero.

The j-th column, $\mathbf{x}$, of A^{-1} satisfies $A\mathbf{x} = \mathbf{e}_j$ since $AA^{-1} = I$ where e_j is the j-th column of I. Solve this system by back substitution, starting from the bottom row.

Every entry of $\mathbf{e}_j$ below row j is 0.

Let $\mathbf{x}$ be the j-th column of A^{-1}, so that $A\mathbf{x} = \mathbf{e}_j$. Then each entry x_k of $\mathbf{x}$ is the (k, j) entry of A^{-1}, and we want to show $x_k = 0$ for every $k > j$.

Assume $j < n$. Every entry of $\mathbf{e}_j$ below row j is 0. The last row of the system $A\mathbf{x} = \mathbf{e}_j$ reads $a_{nn}x_n = 0$ Since $a_{nn} \neq 0$, $x_n = 0$. Thus every column of A^{-1} has zeros below the diagonal, so A^{-1} is upper triangular.

If A is lower triangular, then A^T is upper triangular, so $(A^T)^{-1} = (A^{-1})^T$ is upper triangular. Therefore A^{-1} is lower triangular. ■

Part 2. Suppose A is upper triangular and orthogonal; the lower triangular case is the same. Orthogonality gives $A^TA = I$, so $A^{-1} = A^T$.

By Part 1, A^{-1} is upper triangular. But A^T is lower triangular, because it is the transpose of an upper triangular matrix. Since $A^{-1} = A^T$, this matrix is both upper and lower triangular, so it is diagonal.

Hence A^T is diagonal, and therefore A is diagonal. ■

Question 4: p-norm inequalities (20 points)

Let $\mathbf{x}$ be a real m-vector. The vector p-norms $\|\mathbf{x}\|_p$ are related by various inequalities, often involving the dimension of the vector, i.e. m. For parts (a) and (b), prove the inequality and give an example of a nonzero vector $\mathbf{x}$ for which *equality* is satisfied.

(a) $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$. (7 points)

(b) $\|\mathbf{x}\|_2 \leq \sqrt{m} \cdot \|\mathbf{x}\|_\infty$. (7 points)

(c) Plot a 2D contour of $\|\mathbf{x}\|_1 = 1$; on the same chart also highlight regions where $\|\mathbf{x}\|_2 < 1$, $\|\mathbf{x}\|_2 = 1$ and $\|\mathbf{x}\|_2 > 1$. Describe where the contour intersects $\|\mathbf{x}\|_2 = 1$. (6 points)

Solution:
(a) Start by taking the square of the 1-norm.

$$\|\mathbf{x}\|_1^2 = (|x_1| + |x_2| + \cdots + |x_m|)^2 = \sum_{i=1}^{m} |x_i|^2 + \sum_{i \neq j} |x_i||x_j|$$

The first sum is exactly $\|\mathbf{x}\|_2^2$ and every $|x_i||x_j| \geq 0$. So we get

$$\|\mathbf{x}\|_1^2 \geq \|\mathbf{x}\|_2^2$$

Both norms are nonnegative so we can safely take the square root without the inequality changing. Therefore

$$\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \quad \blacksquare$$

Example: One nonzero entry with all other entries being 0.
Let

$$\mathbf{x} = \begin{bmatrix} 0 \\ -7 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Then

$$\|\mathbf{x}\|_2 = \sqrt{0^2 - 7^2 + 0^2 + \cdots + 0^2} = 7$$

and

$$\|\mathbf{x}\|_1 = |0| + |-7| + |0| + \cdots + |0| = 7$$

(b) Recall that $\|\mathbf{x}\|_\infty = max(|x_1|, |x_2|, ..., |x_m|)$.
Then each entry satisfies $|x_i| \leq \|\mathbf{x}\|_\infty$. And then each $x_i^2 \leq \|\mathbf{x}\|_\infty^2$. There are m $x_i's$ so we sum both sides of the inequality.

$$x_1^2 + x_2^2 + \cdots + x_m^2 \leq \|\mathbf{x}\|_\infty^2 + \|\mathbf{x}\|_\infty^2 + \cdots + \|\mathbf{x}\|_\infty^2 = m\|\mathbf{x}\|_\infty^2$$

We see the left side

$$x_1^2 + x_2^2 + \cdots + x_m^2 = \|\mathbf{x}\|_2^2$$

So we get

$$\|\mathbf{x}\|_2^2 \leq m\|\mathbf{x}\|_\infty^2$$

Since both sides are nonnegative, taking the square root gives

$$\|\mathbf{x}\|_2 \leq \sqrt{m}\|\mathbf{x}\|_\infty \quad \blacksquare$$

Example: All elements in $\mathbf{x}$ are nonzero and have the same absolute value.
Let

$$\mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ \vdots \\ 2 \end{bmatrix}$$

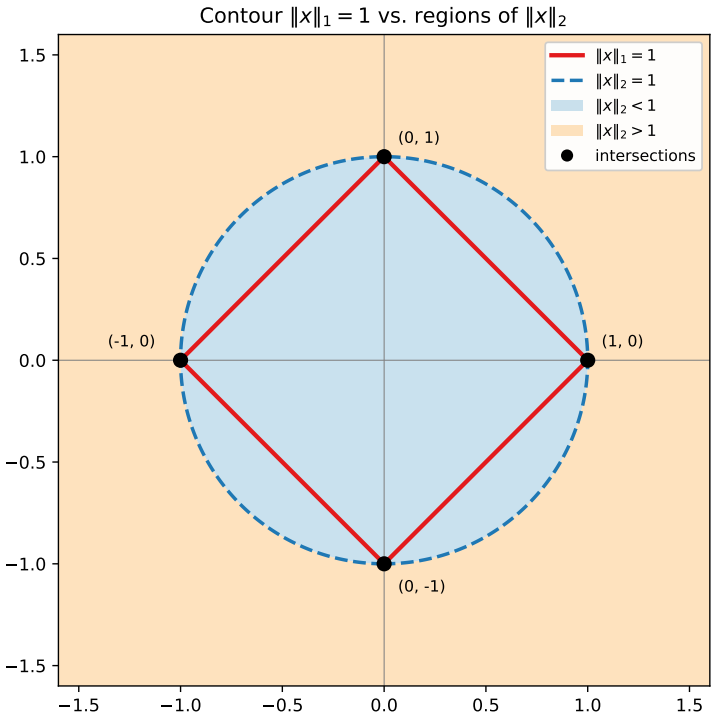
Then

$$\|\mathbf{x}\|_2 = \sqrt{2^2 + 2^2 + \cdots + 2^2} = \sqrt{4 + 4 + \cdots + 4} = \sqrt{4m} = 2\sqrt{m}$$

and

$$\|\mathbf{x}\|_\infty = max(2, 2, ..., 2) = 2 \rightarrow 2\sqrt{m}$$

(c)



The contour $\|x\|_1 = 1$ intersects $\|x\|_2 = 1$ only at the four points $(\pm 1, 0)$ and $(0, \pm 1)$.

Question 5: Basic vector operations (20 points)

Given two 3-dimensional vectors $\mathbf{a}, \mathbf{b}$, and three matrices $\mathbf{A} \in \mathbb{R}^{3\times 2}$, $\mathbf{B} \in \mathbb{R}^{2\times 3}$, $\mathbf{C} \in \mathbb{R}^{3\times 2}$, and scalars β_1, β_2 with the values below:

$$\mathbf{a} = \begin{bmatrix}1\\5\\9\end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix}2\\6\\10\end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix}1 & 4\\2 & 5\\3 & 6\end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix}7 & 8 & 9\\10 & 11 & 12\end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix}1 & 0\\0 & 0\\0 & 1\end{bmatrix}, \quad \beta_1 = 2,\ \beta_2 = 3$$

(a) Compute the following operations by hand and show your work. If an operation is undefined, state why. (15 points)

(i) Vector operations: $\mathbf{a} + \mathbf{b}$, $\beta_1\mathbf{a}$, $\mathbf{a} \circ \mathbf{b}$, $\beta_1\mathbf{a} + \beta_2\mathbf{b}$, where $\circ$ denotes component-wise multiplication. (3 points)

(ii) Matrix operations: $\beta_1\mathbf{A}$, $\mathbf{A} + \mathbf{B}$, $\mathbf{A} + \mathbf{C}$. (3 points)

(iii) Transpose operations: $(\mathbf{BA})^\top$, $\mathbf{A}^\top\mathbf{B}^\top$, $(\mathbf{A}^\top)^\top$, $(\mathbf{A} + \mathbf{C})^\top$. (3 points)

(iv) Inner products and outer product: $\langle \mathbf{a}, \mathbf{b}\rangle$, $\langle \mathbf{b}, \mathbf{a}\rangle$, $\langle \mathbf{a}, \mathbf{a}\rangle$, $\langle \mathbf{b}, \mathbf{b}\rangle$, $\beta_1\langle \mathbf{a}, \mathbf{b}\rangle$, $\langle \beta_1\mathbf{a}, \mathbf{b}\rangle$, $\mathbf{b}\mathbf{a}^\top$. (3 points)

(v) Determinants: $\det(\mathbf{BA})$, $\det(\mathbf{CB})$. (3 points)

(b) Implement all the parts above using Python and show the answers and code. (5 points)

Solution:

(a)

(i)

Vetor operations: $\mathbf{a} + \mathbf{b} = \begin{bmatrix}1+2\\5+6\\9+10\end{bmatrix} = \begin{bmatrix}3\\11\\19\end{bmatrix}$ $\beta_1\mathbf{a} = \begin{bmatrix}\beta_1\\5\beta_1\\9\beta_1\end{bmatrix} = \begin{bmatrix}2\\10\\18\end{bmatrix}$

$\mathbf{a} \circ \mathbf{b} = \begin{bmatrix}1\cdot 2\\5\cdot 6\\9\cdot 10\end{bmatrix} = \begin{bmatrix}2\\30\\90\end{bmatrix}$ $\beta_1\mathbf{a} + \beta_2\mathbf{b} = \begin{bmatrix}\beta_1\\5\beta_1\\9\beta_1\end{bmatrix} + \begin{bmatrix}2\beta_2\\6\beta_2\\10\beta_2\end{bmatrix} = \begin{bmatrix}2\\10\\18\end{bmatrix} + \begin{bmatrix}6\\18\\30\end{bmatrix} = \begin{bmatrix}8\\28\\48\end{bmatrix}$

(ii)

Matrix operations: $\beta_1\mathbf{A} = \begin{bmatrix}\beta_1 & 4\beta_1\\2\beta_1 & 5\beta_1\\3\beta_1 & 6\beta_1\end{bmatrix} = \begin{bmatrix}2 & 8\\4 & 10\\6 & 12\end{bmatrix}$ $\mathbf{A} + \mathbf{B}$ = Undefined since you need the dimension of the matrices to match for matrix addition.

$\mathbf{A} + \mathbf{C} = \begin{bmatrix}1+1 & 4+0\\2+0 & 5+0\\3+0 & 6+1\end{bmatrix} = \begin{bmatrix}2 & 4\\2 & 5\\3 & 7\end{bmatrix}$

(iii)

Transpose operations: $(\mathbf{BA})^\top = \begin{bmatrix}7\cdot 1+8\cdot 2+9\cdot 3 & 7\cdot 4+8\cdot 5+9\cdot 6\\10\cdot 1+11\cdot 2+12\cdot 3 & 10\cdot 4+11\cdot 5+12\cdot 6\end{bmatrix}^\top = \begin{bmatrix}50 & 122\\68 & 167\end{bmatrix}^\top = \begin{bmatrix}50 & 68\\122 & 167\end{bmatrix}$

$\mathbf{A}^\top\mathbf{B}^\top = \begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}\begin{bmatrix}7 & 10\\8 & 11\\9 & 12\end{bmatrix} = \begin{bmatrix}1\cdot 7+2\cdot 8+3\cdot 9 & 1\cdot 10+2\cdot 11+3\cdot 12\\4\cdot 7+5\cdot 8+6\cdot 9 & 4\cdot 10+5\cdot 11+6\cdot 12\end{bmatrix} = \begin{bmatrix}50 & 68\\122 & 167\end{bmatrix}$

$(\mathbf{A}^\top)^\top = \begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}^\top = \begin{bmatrix}1 & 4\\2 & 5\\3 & 6\end{bmatrix}$ $(\mathbf{A} + \mathbf{C})^\top = \begin{bmatrix}1+1 & 4+0\\2+0 & 5+0\\3+0 & 6+1\end{bmatrix}^\top = \begin{bmatrix}2 & 4\\2 & 5\\3 & 7\end{bmatrix}^\top = \begin{bmatrix}2 & 2 & 3\\4 & 5 & 7\end{bmatrix}$

(iv)
Inner and outer product operations: **(b)**

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import numpy as np
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